Beginner Course on Microbial Genomics: Applications & Research (2026)

# Module 1.3

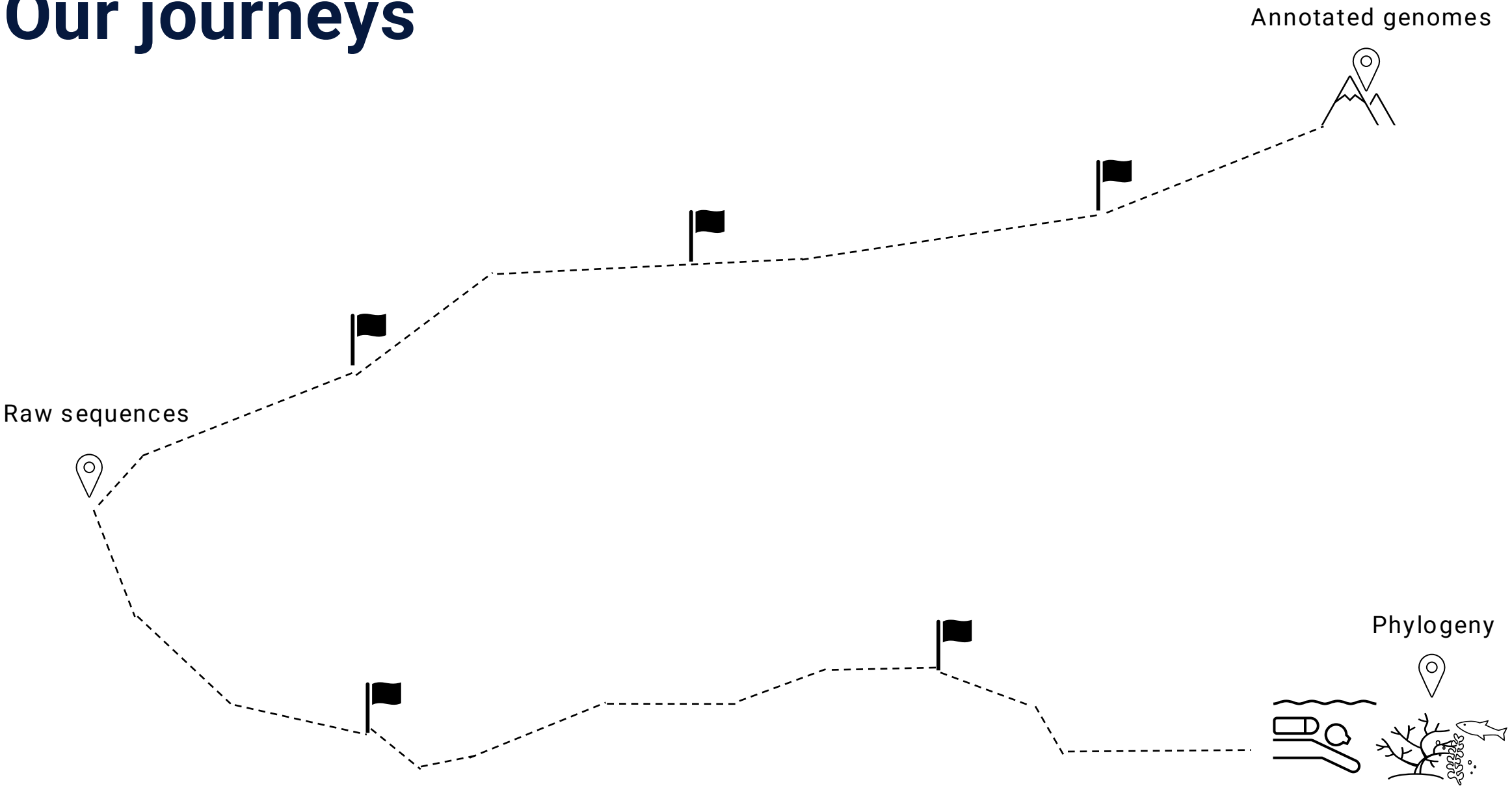
# From raw sequencing data to assemblies

## INFORMATION

**Lecturer**

Mai Thu Si Nguyen

# Our journeys



# Study case -
# a nosocomial *K. pneumoniae* outbreak

Outbreaks

Open Access

Hospital outbreak sustained by *Klebsiella pneumoniae* sequence type 147 co-producing NDM-1 and OXA-48, Rome, Italy, February to March 2025: molecular tracing and control measures

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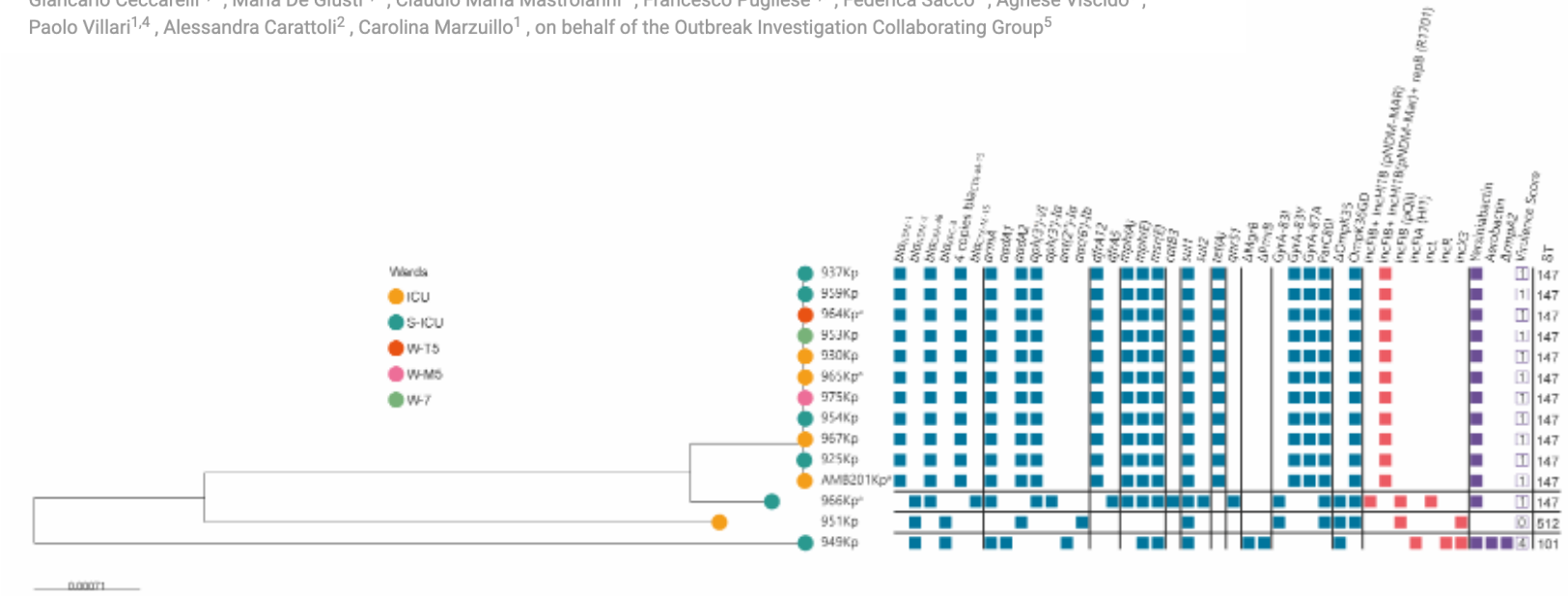
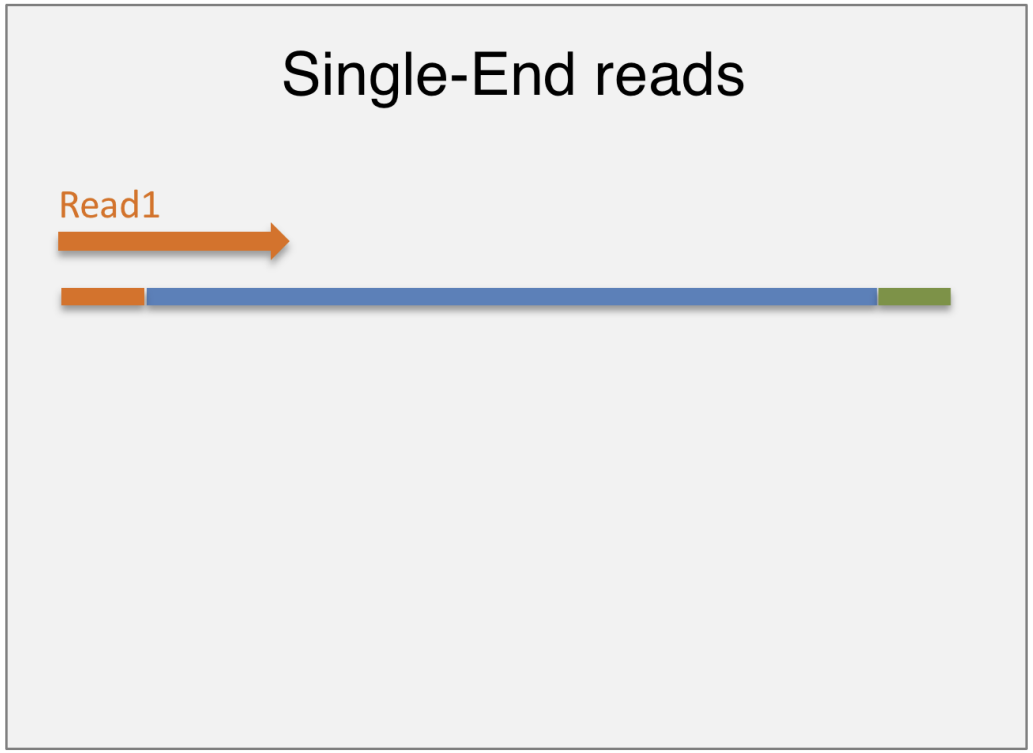
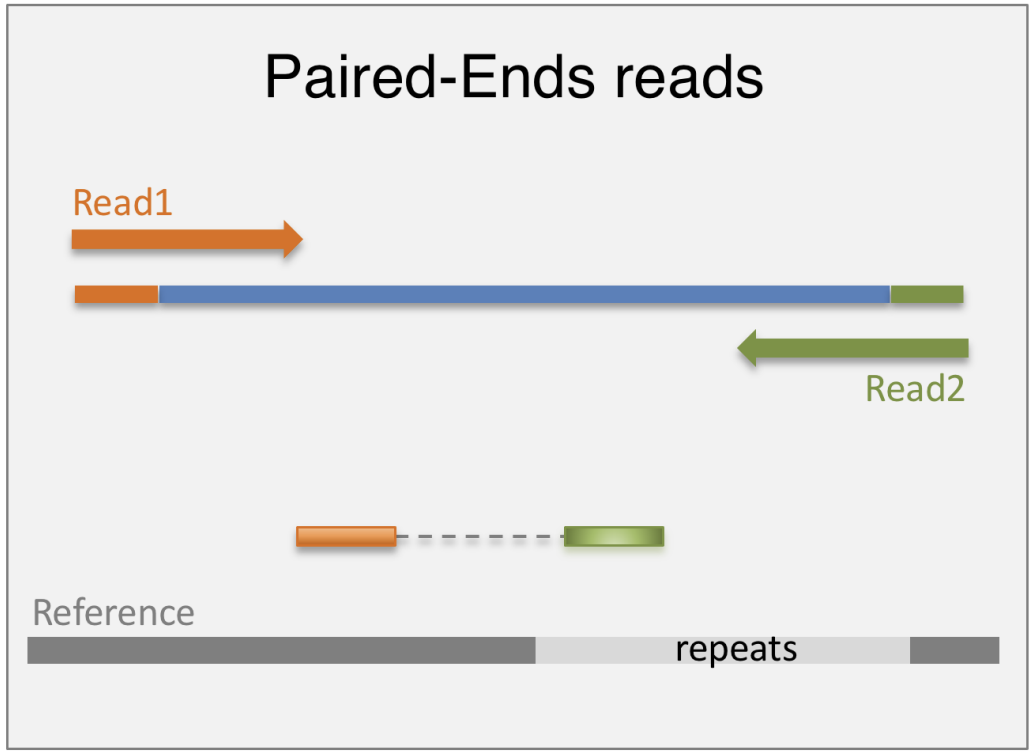
Valerio Capitani[1,*], Mariateresa Ceparano[1,*], Annalisa Rosso[1], Guido Antonelli[2,3], Maria Augurusa[4], Valentina Baccolini[1], Giancarlo Ceccarelli[1,4], Maria De Giusti[1,4], Claudio Maria Mastroianni[1], Francesco Pugliese[4,5], Federica Sacco[3], Agnese Viscido[3], Paolo Villari[1,4], Alessandra Carattoli[2], Carolina Marzuillo[1], on behalf of the Outbreak Investigation Collaborating Group[5]



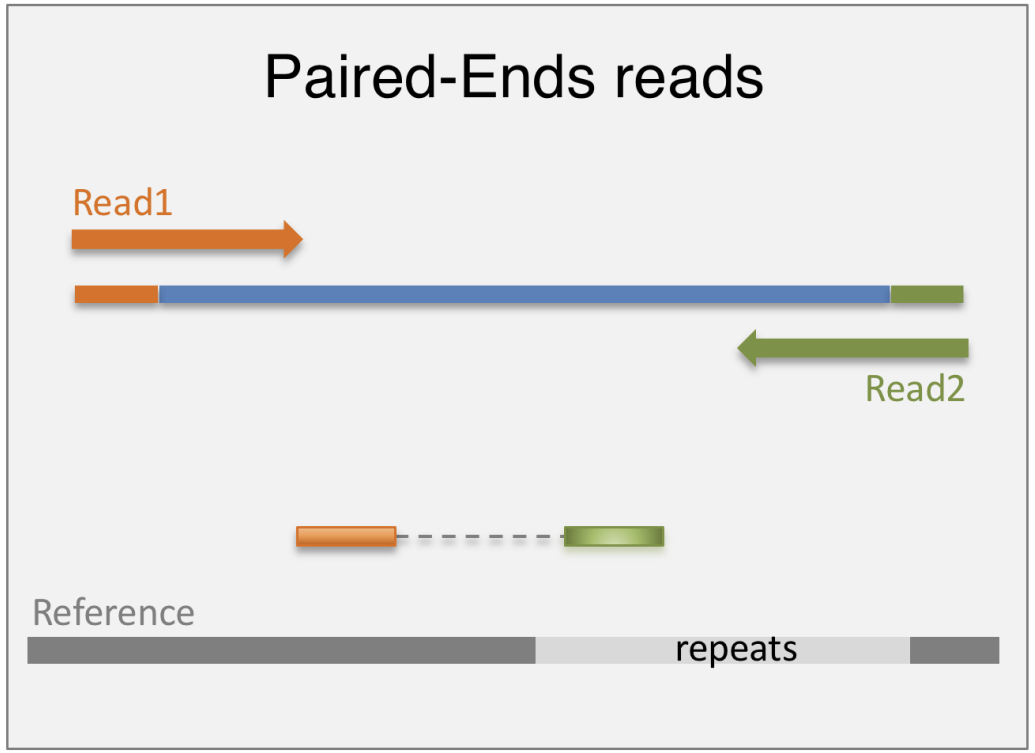
# Raw sequencing data



# Raw sequencing data



# Raw sequencing data

```
.
├── PRJNA1260529.metadata.tsv
├── raw_longread
├── raw_shortread
├── samples_shortread.tsv
├── script.sh
```

Let's take a look at the files in the `raw_shortread` directory:

```
ls raw_shortread/
```

You shall see a directory structure that looks like this:

```
├── raw_shortread
│   ├── 925Kp_1.fastq.gz
│   ├── 925Kp_2.fastq.gz
│   ├── 930Kp_1.fastq.gz
│   ├── 930Kp_2.fastq.gz
│   ├── ...
```

Try to answer the following questions:

- How many files are there?
- What are the sizes of these files?
- What do the file names tell you?

| | | |
|---|---|---|
| Application or product: | Whole-Genome Sequencing | |
| Coverage: explain this | 50 | × |
| Duplicates: explain this | 2 | % |
| Genome or region size (in million bases): | 5.5 | Mb |
| Total read length (e.g. 200 for 2x100): | 300 | cycles |

Output Required: 280,612,245 bases

**Benchtop Sequencers**

- [ ] iSeq
- [ ] MiniSeq
- [x] MiSeq / MiSeq Dx in RUO mode
- [ ] NextSeq 500/550

**Production-Scale Sequencers**

- [ ] NextSeq 500/550
- [ ] NovaSeq 6000
- [ ] HiSeq 3000/4000
- [ ] HiSeq 1500/2500 Rapid Run
- [x] HiSeq 1500/2500 High Output
- [ ] NextSeq 1000 Sequencing System
- [ ] NextSeq 2000 Sequencing System

Submit

| | MiSeq | MiSeq |
|---|---|---|
| Run type | v3 Reagents | v2 Reagents |
| Clusters | 25,000,000 per flow cell | 15,000,000 per flow cell |
| Output per unit (flow cell or lane) | 7,500,000,000 per flow cell | 4,500,000,000 per flow cell |
| Exceeds maximum read length? | Does not exceed maximum (2x300) | Does not exceed maximum (2x250) |
| Number of units per sample (flow cell or lane) | 0.037 flow cells | 0.062 flow cells |
| Samples per unit (flow cell or lane) | 26/flow cell | 16/flow cell |

https://support.illumina.com/downloads/sequencing_coverage_calculator.html

# FASTQ files

```
@SRR33475255.1 VH01637:40:AAGVVC2M5:1:1101:22246:1000 length=301
AAGTACCTCAACGGCCATTCCGATGTCGTCGCCGGCCTGGCGGTGGTTGGCGATAACTCCGGGCTCGCGGAAAAGCTTGGCTATCTGCAGAATGCGGTAGGCG
GCGTGCTTGACCCGTTCAGCAGCTTCCTGACGCTGCGCGGCATCCGTACCCTGGCGCTGCGGATGGAGCGCCACAGCGCGAACGCGCTGCAGCTCGCCGAATG
GCTGGAACAGCAGCCGGAAGTGGAGCGAGTCTGGTTTCCCTGGCTGGCCTCGCACCCGCATCATCAGCTGGCGCGCCAGCAAATGGCCCTGCCCG
+
CCCCCC*CCCCCCCCCCCCC5CCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCC5C5CCCCCCCCCCCCCCCCCCCCCC5CCCCCCCCCCCCCCCCCCCCCC
CCCCCCCCCCCCCC*CCCCCCCCCCC5CCCCCCCC5CCCCC5CCCCCCC5CCCCCCCCCCCCCCCC*CCCCCCCCCCCCCC5*CCCC5CCCCCC5CCCC*CCCCCCCCC
CCCCCCCCCCCCCC*CCC*C5CCCCCCCCCCCCC*CCCCCCCCCCCCC5CCCCCCCCC*CC*CCC55*5CC55*5C*C555C5CCC5C**
@SRR33475255.2 VH01637:40:AAGVVC2M5:1:1101:28797:1000 length=199
ATGTTGACCTGTAAGAACGAAAACACTGTTCTTTCCTGTGTGTCTTCCATTTCTTAGAACAACGTGCATATCTATTCTTGCATTTAAAGTCGCTTCGTCAATCCCCTAT
ATTTCTAGGCGGGTATCTTTTAATGCAGATGGGAATACTCATCTCATAAACAGTGAATAATTATAAAGATGAAATCACACGGTTCTATTT
+
*CCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCC5CCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCC
CCCC5CCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCC5CCCCCCCCC*CCCCC5CCC*5CCCCCCC5CCCCCCC5CCC*C55CCCCC5
@SRR33475255.3 VH01637:40:AAGVVC2M5:1:1101:32244:1000 length=295
CGGTAACTGATGCCGTATTTGCAGTACCAGCGTACGGCCCACAGAATGATGTCACGCTGAAAATGCCGGCCTTTGAATGGGTTCATGTGCAGCTCCATCAGCAAA
AGGGGATGATAAGTTTATCACCACCGACTATTTGCAACAGTGCCCATCTCCGGGTTTATTATATTCGGGCGCATTTGCGCGCGGCAGCTGCTGACTGTTCAGTCTAT
GTACGCGGAGCCGTTCATGGAAATATTCGCGCAGGTGTTCAGGCTGTTCCCGTTCAACAAGTTCCGGCACGACGGGCTTGTTC
+
*CCC5CCCCCCCCCCCCCCCCCCCCCCCCCCC5CCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCC5CCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCC
CCCCCC5CCCCCCCCCCCCCCCCCCCCCCCCC5CCCCCCC5CCCCCCCCCCCCCCC5CCCC*CCCC*CCCCC5CCC*CCCCC5CCCCCCCCCCCCCCCCCCCCCCCCCC5C
CCCCCCCCCCCCCCC*CCCCCCC5CCCCCCCCCCCCCCCCCCCCCCCCCCCCC5CCCCCCCCCC5C5C5CCCCCCCCCCCCCC
```

*How is FASTQ format different from FASTA?*

# Quality Score

```
+SEQ_ID
!''*((((***+))%%%++)(%%%%).1**
```

A quality value $Q$ is an integer representation of the probability $p$ that the corresponding base call is incorrect.

$$Q = -10 \log_{10} P \quad \Longrightarrow \quad P = 10^{\frac{-Q}{10}}$$

| Phred Quality Score | Probability of incorrect base call | Base call accuracy |
|---|---|---|
| 10 | 1 in 10 | 90% |
| 20 | 1 in 100 | 99% |
| 30 | 1 in 1000 | 99.9% |
| 40 | 1 in 10000 | 99.99% |
| 50 | 1 in 100000 | 99.999% |

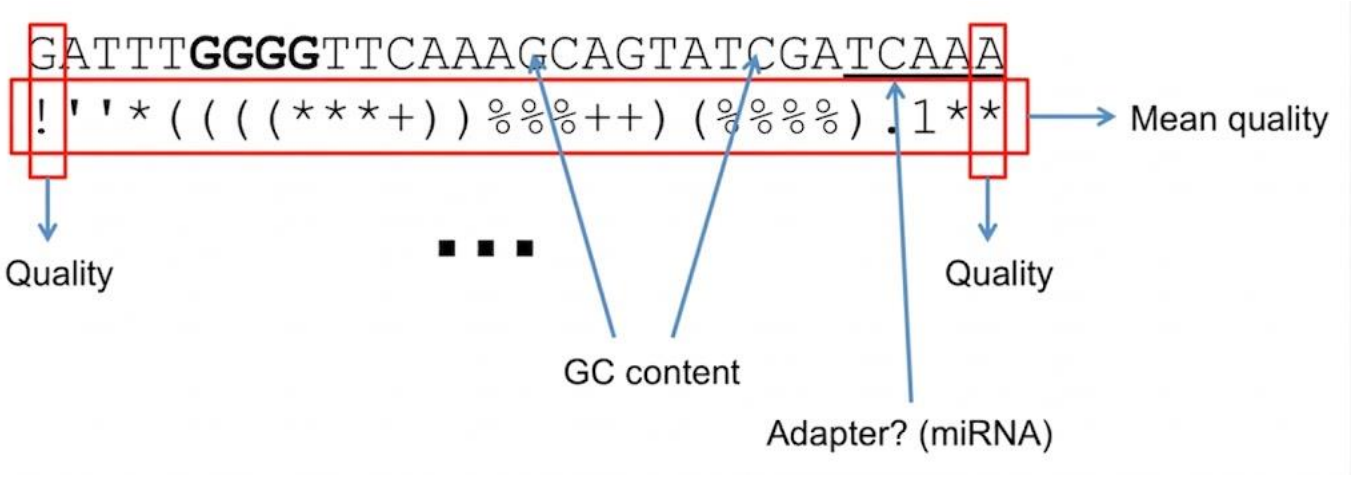
| *Symbol* | *ASCII Code* | *Q-Score* |
|---|---|---|
| *!* | *33* | *0* |
| *"* | *34* | *1* |
| *#* | *35* | *2* |
| *$* | *36* | *3* |
| *%* | *37* | *4* |
| *&* | *38* | *5* |
| *'* | *39* | *6* |
| *(* | *40* | *7* |
| *)* | *41* | *8* |
| *** | *42* | *9* |
| *+* | *43* | *10* |
| *,* | *44* | *11* |
| *-* | *45* | *12* |
| *.* | *46* | *13* |
| */* | *47* | *14* |
| *0* | *48* | *15* |
| *1* | *49* | *16* |
| *2* | *50* | *17* |
| *3* | *51* | *18* |
| *4* | *52* | *19* |
| *5* | *53* | *20* |

| Symbol | ASCII Code | Q-Score |
|---|---|---|
| *6* | 54 | 21 |
| *7* | 55 | 22 |
| *8* | 56 | 23 |
| *9* | 57 | 24 |
| *:* | 58 | 25 |
| *;* | 59 | 26 |
| *<* | 60 | 27 |
| *=* | 61 | 28 |
| *>* | 62 | 29 |
| *?* | 63 | 30 |
| *@* | 64 | 31 |
| *A* | 65 | 32 |
| *B* | 66 | 33 |
| *C* | 67 | 34 |
| *D* | 68 | 35 |
| *E* | 69 | 36 |
| *F* | 70 | 37 |
| *G* | 71 | 38 |
| *H* | 72 | 39 |
| *I* | 73 | 40 |

https://learn.gencore.bio.nyu.edu/ngs-file-formats/quality-scores/
https://support.illumina.com/help/BaseSpace_Sequence_Hub_OLH_009008_2/Source/Informatics/BS/QualityScoreEncoding_swBS.htm

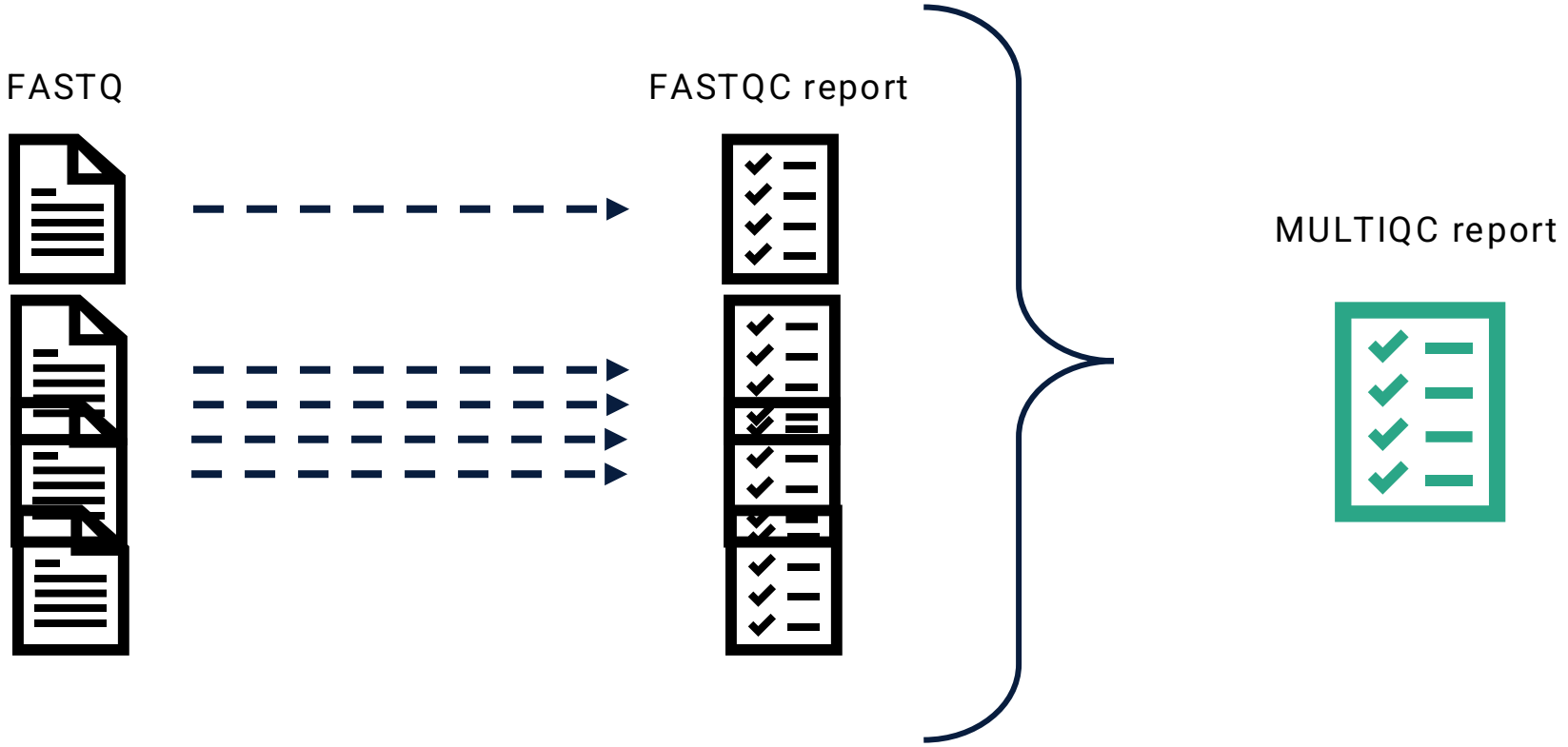
# Reads Quality Control (QC)

- ❑ Reads Coverage (estimates) ← Number of reads
- ❑ Base Quality (Phred scores)
- ❑ GC Content
- ❑ Per Base N Content
- ❑ Adapter Contamination

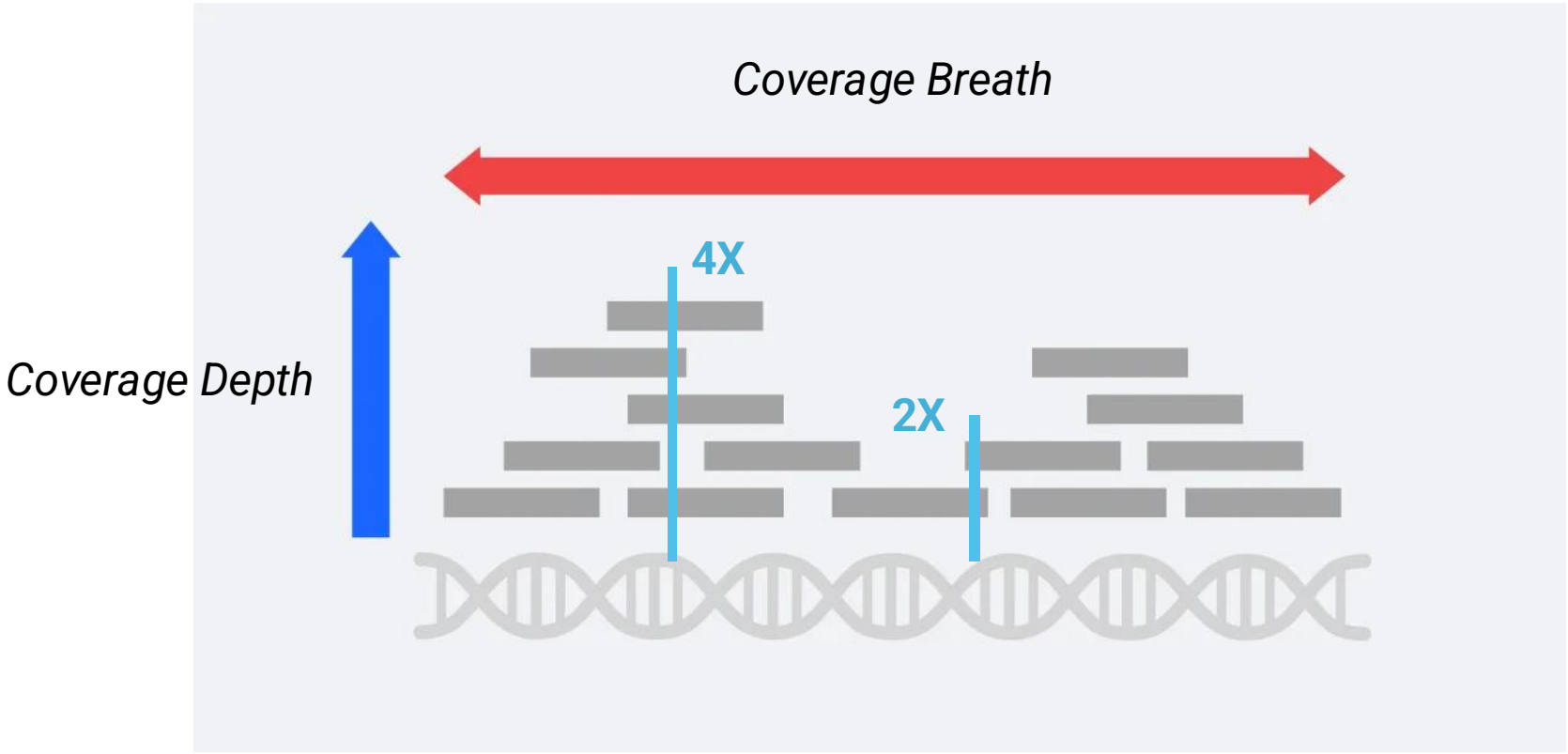



https://learn.gencore.bio.nyu.edu/ngs-file-formats/quality-scores/

# QC on our data



# Read Coverage




- ❑ Number of reads
- ❑ Read length

# Reads Quality Control (QC)

- **AVERAGE** Reads Coverage (estimates) ← Number of reads

$$\frac{\text{Total number of reads} \times \text{Read length}}{\text{Chromosome length}}$$

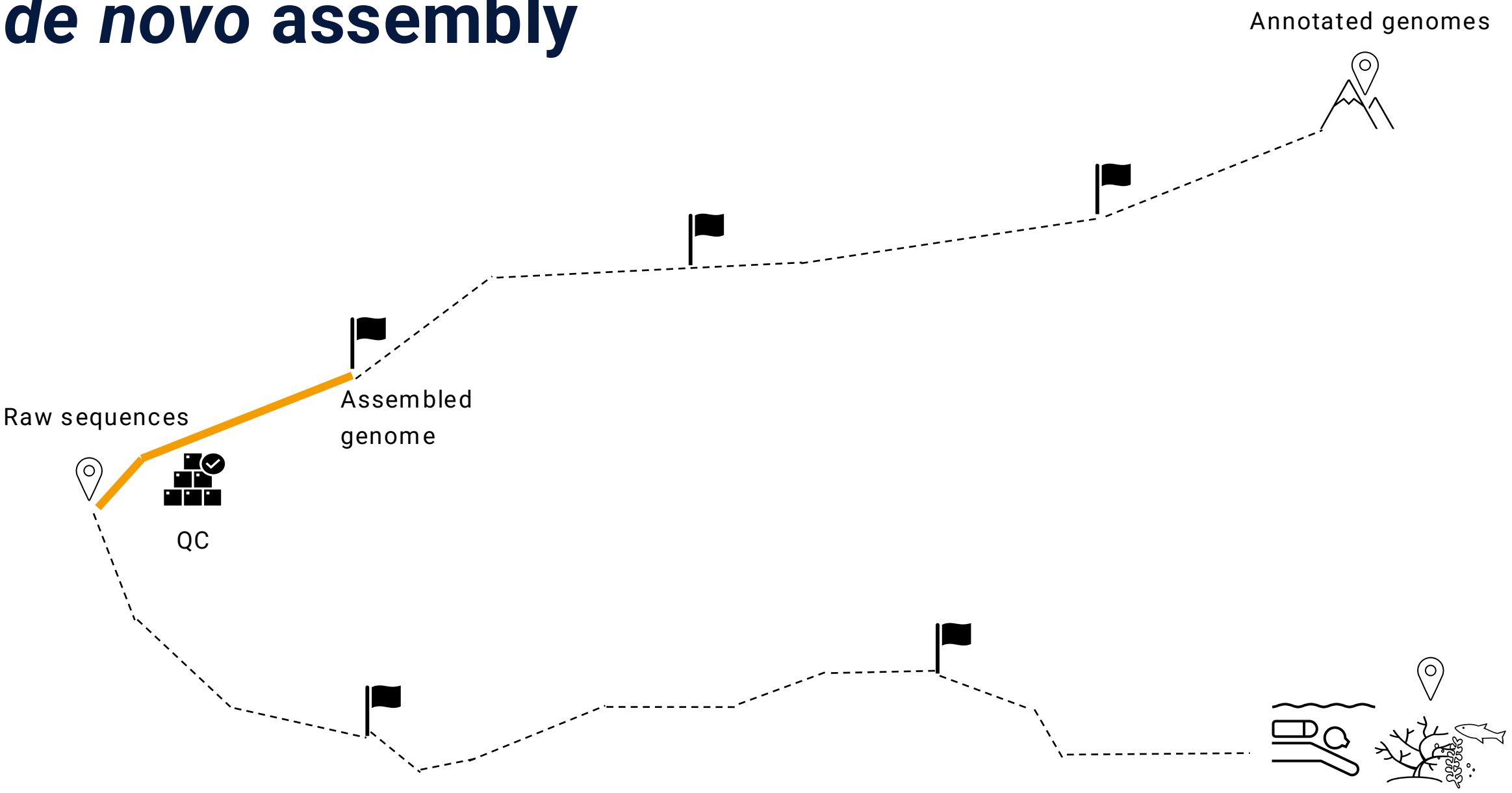
# QC on our data

**Making decision**

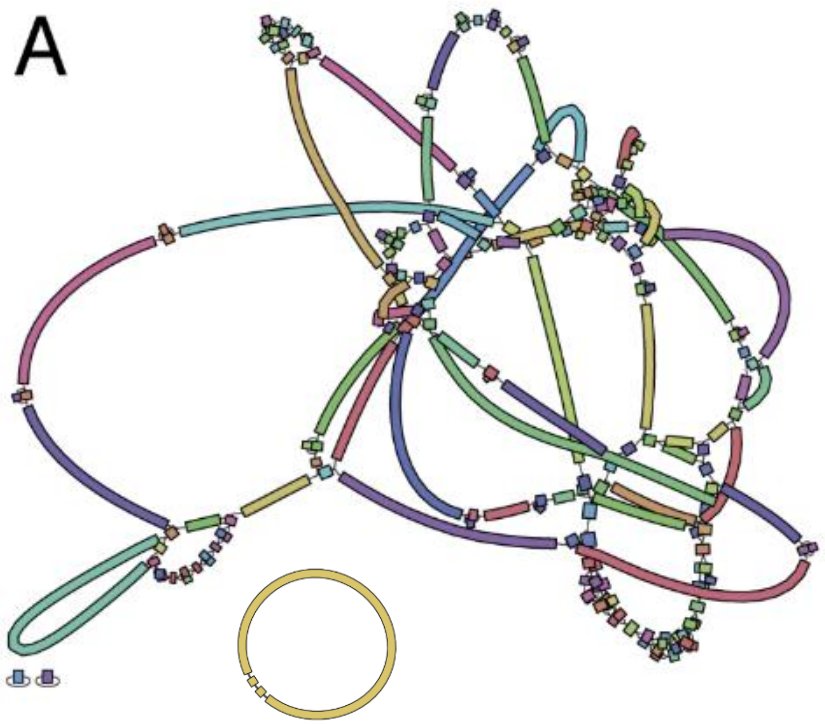
Based on the MultiQC report, do you want to improve the raw reads quality?

If yes, what specific modification to the raw reads would you like to perform? If no, then why?

# *de novo* assembly



# What is *de novo* assembly ?



# Assembly by Unicycler

```
unicycler -1 $forward -2 $reverse -o $outdir --threads 4
```

where

- `-1` is the forward read file
- `-2` is the reverse read file
- `-o` is the output directory
- `--threads` is the number of threads to use, higher number of threads will speed up the assembly process but will also consume more computing resources.

# Assembly by Unicycler

```
Just contigs:                Assembly graph:

TCGAAACTTGACGCGAGTCGC                            CTTGTTTA
TGCTACTGCTTGATGATGCGG                           /        \
TGTCCATT                     TCGAAACTTGACGCGAGTCGC          TGCTACTGCTTGATGATGCGG
CTTGTTTA                                        \        /
                                                 TGTCCATT
```

WRAP-UP

# Day 1: Meet the Genomes

- ✓ Understand applications of WGS in clinical microbiology
- ✓ Study public genomes on NCBI database - Inspect gene regions
- ✓ Practice on raw sequencing data:
  - Quality control
  - *De novo* assembly

# THE END.

*Tranh Họa sĩ và Người mẫu III,*
*Pablo Picasso,*
*1970.*

**Day 1: Meet the Genomes**